Question 1

Platform and prior run. The selected MLOps workflow is a full-code Python stack: FLAML AutoML for the primary search, with Feast for feature storage and MLflow for experiment tracking. H2O AutoML is the requested second run. The workflow is full-code because scripts call the Python APIs directly; it does not use a no-code modeling interface, so the no-code screenshot instruction does not apply.

All experiment results below come from the retained October 18, 2026 run. The FLAML search scores are the original trial validation scores. FLAML speed values were measured November 3 by refitting each logged trial configuration on the original training split and timing it on the original common validation batch. H2O leaderboard scores and speed values come from its saved October 18 run.

The source archive is `athletes.csv.zip`, SHA-256 (64 hex characters):

```
4a7a2701564ca0c7d2bc5016964aab69
7abacf25ac74abaa28deda8c6689dd6
```

It contains 423,006 rows. The prepared cohort has 70,325 rows, split by gender with seed 42 into 49,227 train, 10,549 validation, and 10,549 test rows. The A3 AutoML prompt does not specify a target formula; for comparison with Part A, I retain `total_lift = candj + snatch + deadlift + backsq`. The source supplies no unit metadata. The lift components and identifiers are not predictors.

Table 1: Validation-selected models and measured prediction speed

Run / feature mode	Selected model	Validation RMSE	Median ms / 1,000 rows
Part A prior baseline, v1	Random forest, 80 trees / depth 8 / leaf 5	165.537	14.76
FLAML, all 10 features	LightGBM, trial 23	165.369	1.609
FLAML, selected top 3	LightGBM, trial 23	172.516	1.493
H2O AutoML, all 10 features	GBM_2	165.263	215.76
H2O AutoML, selected top 3	GBM_2	176.336	224.29

The all-feature FLAML and H2O winners are close to the prior Part A forest on validation RMSE. The FLAML measurements are local Python prediction calls with wrapper and dataframe overhead. H2O measurements include the client request, REST/JVM serialization, and H2OFrame conversion, so cross-platform speed is not a pure algorithm comparison.

Question 2

Insights and top five features. Females are 27.7% of the cohort and males 72.3%. The prior-work target median is 1,025 in unlabeled raw units. Both AutoML runs rank `is_female` and `weight` as the two strongest predictors. Their next-ranked features differ: FLAML selects `height`, while H2O selects `weight_squared`.

Table 2: Validation permutation importance, mean RMSE increase after permutation

Rank	FLAML full 10	Increase	H2O all 10	Increase
1	`is_female`	102.9	`is_female`	98.6
2	`weight`	81.2	`weight`	40.8
3	`height`	15.0	`weight_squared`	14.2
4	`age_x_height`	7.7	`age_x_height`	6.0
5	`age_x_weight`	3.8	`height`	4.9

Each feature-importance analysis used a 2,000-row validation sample, three permutations per feature, and no test data. The values are specific to their platform and source model; they should not be compared as a shared scale.

Question 3

FLAML validation-score and speed rankings. The top-three validation models are ranked by the original October 18 trial losses. Each trial ID identifies a unique learner configuration. The speed columns show November 3 measurements after refitting those logged configurations on the original training split. All models used the same 1,000 validation IDs, one warm-up, and five measured predictions.

Table 3: FLAML, all 10 features

Val. rank	Trial / learner	RMSE	Speed rank	Trial / learner	RMSE	ms / 1,000
1	23 / `lgbm`	165.369	1	24 / `rf`	166.509	1.135
2	18 / `rf`	166.223	2	20 / `lgbm`	168.676	1.166
3	24 / `rf`	166.509	3	18 / `rf`	166.223	1.182

Table 4: FLAML, selected top 3 features: `is_female`, `weight`, and `height`

Val. rank	Trial / learner	RMSE	Speed rank	Trial / learner	RMSE	ms / 1,000
1	23 / `lgbm`	172.516	1	8 / `lgbm`	207.756	0.784
2	18 / `rf`	172.765	2	1 / `extra_tree`	188.227	0.832
3	27 / `rf`	172.773	3	7 / `extra_tree`	187.434	0.835

The top-three validation list includes distinct trial configurations, not only one retained configuration per learner family. The October 18 validation ranking is preserved; all 60 November 3 refits reproduced their corresponding logged validation RMSEs.

Question 4

H2O AutoML repeat. H2O ranks its six actual leaderboard models in each feature mode. The top-three feature set selected from all-feature validation importance is `is_female`, `weight`, and `weight_squared`. H2O model labels below use the leaderboard suffixes; the full IDs are retained in the source outputs. All-feature run ID: `AutoML_1_20261018_215612`; top-three run ID: `AutoML_2_20261018_215643`.

Table 5: H2O AutoML, all 10 features

Val. rank	Model	RMSE	Speed rank	Model	RMSE	ms / 1,000
1	GBM_2	165.263	1	GLM_1	174.740	5.85
2	GBM_3	165.810	2	GBM_2	165.263	215.76
3	GBM_1	166.202	3	GBM_3	165.810	217.90

Table 6: H2O AutoML, selected top 3 features

Val. rank	Model	RMSE	Speed rank	Model	RMSE	ms / 1,000
1	GBM_2	176.336	1	GLM_1	181.249	4.83
2	GBM_1	176.370	2	GBM_4	176.554	216.62
3	GBM_3	176.401	3	DRF_1	176.825	222.66

Question 5

Comparison with Part A. The full-feature FLAML trial 23 has validation RMSE 165.369, compared with 165.537 for the selected Part A forest; its November 3 refit measures 1.609 ms per 1,000 predictions. The full-feature H2O winner has validation RMSE 165.263 and measures 215.76 ms per 1,000 predictions in the saved October 18 H2O client timing. Top-three feature selection raises validation RMSE to 172.516 for FLAML and 176.336 for H2O. The fastest top-three FLAML trial measures 0.784 ms per 1,000 predictions but has RMSE 207.756. These timing figures reflect the stated local call paths, including framework overhead.

The accompanying code includes the original training workflow and the candidate refit/timing script. Working code: GitHub repository.